



Brendon Mutema

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Electrical & Computer Engineer | Python, 5G/Wi-Fi Coexistence & Applied ML | UCT Engineering Graduate

University of Cape Town

City of Cape Town, Western Cape, South Africa · Contact info

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About

As a recent Electrical and Computer Engineering graduate and an incoming Master's student at the University of Cape Town, I am passionate about solving complex challenges at the intersection of wireless communications and system engineering. My academic journey has provided me with a robust foundation in software development, system analysis, and practical hardware implementation.

My final year thesis on 5G/Wi-Fi coexistence allowed me to develop a discrete-event simulator from scratch in Python, honing my skills in system modeling, performance optimization, and data-driven analysis. This experience, combined with my upcoming Master's research on 5G/satellite coexistence for ICASA, demonstrates my commitment to tackling real-world spectrum management problems.

I am a proactive and resilient learner always eager to master new technologies. I thrive in collaborative environments and am now seeking graduate or internship opportunities where I can contribute my technical skills and grow as an engineer. Let's connect!

Top skills
Python (Programming Language) • Wireless Communications • Machine Learning • 5G NR and Wi-Fi Coexistence

Activity

119 followers

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Electrical & Computer Engineer | Python, ...
2mo

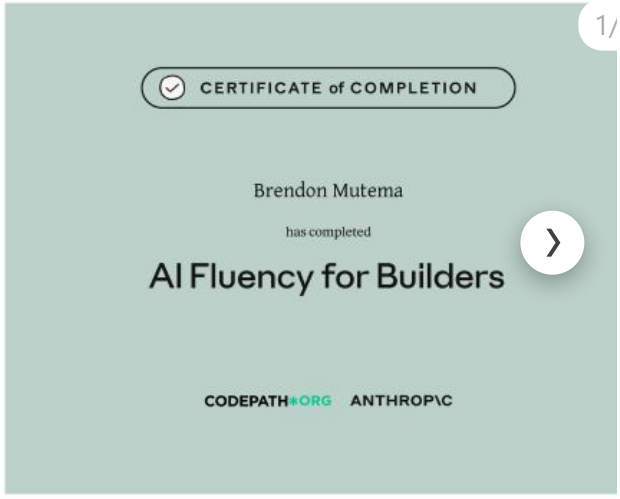
Full technical stack complete: **20/20 certifications cleared**. 🚀 ... more



4

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Electrical & Computer Engineer | Python, ...
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New Milestone: 10 technical certifications complet 🚀 ... more



1

162 impressions View analytics 193 impressions View analy

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Experience

Machine Learning Internship

FlyRank AI · Internship
Jul 2026 - Sep 2026 · 3 mos
Pretoria, Gauteng, South Africa · Remote

Capstone finding: a model scoring 0.94 precision@50 on a naive split fell to 0.22 on unseen clients, below the 0.279 base rate. Completed as a Machine Learning Engineering Intern at FlyRank AI, alongside the General AI Fluency Programme.

Machine Learning Capstone
Investigated whether a model could outperform random selection when prioritizing declining content for review, using 17.7M rows of search data across 349,411 content items and 59 clients.

• Defined a data contract and conducted data leakage checks using Python.
• Compared a Random Forest and an age-and-visibility scoring rule against random selection using a client-grouped evaluation split.
• Found that precision@50 fell from 0.94 on a naive split to 0.22 on unseen clients, below the 0.279 base rate. The rule-based baseline scored 0.10.
• Recommended a human-review playbook rather than automated prioritization because neither approach generalized adequately.

Case study: betmutema.github.io/ml-engineering-internship
Code: github.com/betmutema/ml-engineering-internship

General AI Fluency Programme
Completed the 10-week Programme and delivered an accepted Impact Project.

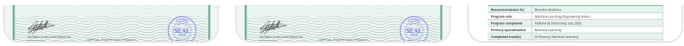
• Built and launched my engineering portfolio using HTML, CSS and GitHub Pages, featuring case studies on 5G NR-U and Wi-Fi coexistence and TOPSIS-based radio access technology selection.
• Built a Weekly Review Assistant using a Claude Project connected to Gmail and Google Calendar. Designed it for read-only operation and tested seven scenarios, including tool failure and irrelevant keyword matches.
• Conducted mobile and robustness testing on the portfolio and documented AI contributions and independent decisions using the 4D framework: Delegation, Description, Discernment and Diligence.

Portfolio: betmutema.github.io
Portfolio code: github.com/betmutema/betmutema.github.io
AI assistant write-up: github.com/betmutema/flyrank-ai-fluency-final-package









Machine Learning, Python (Programming Language) and +1 skill

Tutor

MindShift Learning · Part-time

Mar 2026 - Aug 2026 · 6 mos

City of Cape Town, Western Cape, South Africa · Remote

Leveraging my background in Electrical and Computer Engineering to provide specialized academic support in STEM subjects, including Mathematics and Engineering principles.

Key responsibilities include:
• Translating complex technical concepts into accessible, structured learning materials.
• Mentoring students in advanced problem-solving methodologies and analytical thinking.
• Developing technical communication skills essential for bridging the gap between high-level theory and practical application.

Technical Communication, Analytical Skills and +2 skills

Engineering Intern

Mamoshe Investment Group (Pty) Ltd · Internship

Aug 2025 - Sep 2025 · 2 mos

Polokwane, Limpopo, South Africa · On-site

Completed six weeks of practical training (EEE3000X) contributing to the CCTV installation project at Waterberg TVET College. The role involved hands-on implementation and technical support, including installing, terminating, and testing CAT6 network cables according to the T568B standard to ensure reliable connectivity. Assisted senior engineers with configuring network switches and implementing VLANs to isolate and secure CCTV traffic from the main data network. Applied IEEE 802.3at (PoE+) principles to verify power delivery to IP cameras, considering switch power budgets and cable length limitations. Also supported prototype testing and provided general IT system assistance as required by the project team.



MIG Letter - Mutema BET.pdf
A letter from Mamoshe Investment Group confirming the successful completion of my six-week vacation work (EEE3000X). The letter outlines my role and ...

Wireless Communications, Technical Support and +3 skills

Vacation Work Trainee

Cape Peninsula University of Technology · Internship

Dec 2021 · 1 mo

City of Cape Town, Western Cape, South Africa · On-site

Completed a two-week intensive training program at Cape Peninsula University of Technology, covering Welding, Pneumatics, Metrology, Fitting and Turning, Robotics, and Basic Engineering Hand Skills. During the vacation work training program, I applied theoretical knowledge to practical tasks and developed hands-on technical skills that strengthened my understanding of industrial processes and engineering workshop practices.



Brendon E T Mutema - Certificate of Training.pdf
Official Certificate of Training awarded to Brendon E. T. Mutema by Cape Peninsula University of Technology, confirming successful completion of the ...

Education

University of Cape Town
Bachelor of Science - BSc, Electrical & Computer Engineering
Jan 2019 – Jun 2025

Activities and societies: Member of the UCT IEEE (Institute of Electrical and Electronics Engineers) Student Branch

Completed a comprehensive NQF Level 8 degree with a curriculum focused on software development, embedded systems, electronics, and communication systems. The program emphasized project-based learning and a systematic approach to engineering problem-solving.

Final Year Thesis: Coexistence Strategy for 5G New Radio and Wi-Fi
My capstone project involved designing and validating an "Optimized Gap Mode" strategy to mitigate interference in shared spectrum. Key achievements included:
• Developed a discrete-event simulator from scratch using Python (SimPy) to model complex network interactions.
• Achieved near-perfect fairness (Jain's Fairness Index > 0.97) and balanced channel occupancy in simulations, demonstrating a significant improvement over baseline models.

Project Management, AutoCAD and +16 skills




Pamushana Secondary School
Zimbabwe General Certificate of Education at Advanced Level, Physical Sciences and Mathematics
Feb 2017 – Nov 2018

Grade: Achieved three A's and two B's

Completed a rigorous pre-university curriculum based on the Cambridge A-Level system, focused on advanced-level science and mathematics. This qualification provided a strong analytical and problem-solving foundation for my subsequent engineering studies at the University of Cape Town.

Final subjects and results:
• Computer Science (A)
• Chemistry (A)
• Statistics (A)
• Physics (B)
• Pure Mathematics (B)

Analytical Skills



Brendon Mutema - Matriculatin Certification & A'Level Certificate.pdf
Official documentation containing the Matriculation Certificate issued by Universities South Africa (USAf) and the Zimbabwe General Certificate of ...

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Licenses & certifications (20)

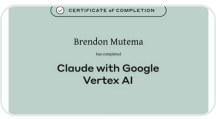
Claude with Google Cloud's Vertex AI

Anthropic

Issued Aug 2026

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Vertex AI



Claude on Google Cloud - Certificate of Completion

Technical certification (93 lessons) on integrating and scaling AI models within the Google Cloud ecosystem. Expertise in Vertex AI configuration, systematic ...

Claude with Amazon Bedrock

Anthropic

Issued Aug 2026

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Retrieval-Augmented Generation (RAG) and Amazon Web Services (AWS)



Claude with Amazon Bedrock - Certificate of Completion

Technical certification (83 lessons) on architecting and deploying AI solutions within the AWS ecosystem. Validated expertise in AWS Bedrock API integration...

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Projects

Final Year Thesis: Coexistence Strategy for 5G New Radio

and Wi-Fi

Jan 2025 – Jun 2025

My capstone project for my Bachelor of Science in Engineering. I designed and validated an "Optimized Gap Mode" strategy to mitigate interference and ensure fair coexistence between 5G NR-U and Wi-Fi networks in shared spectrum.
To achieve this, I developed a discrete-event simulator from scratch using Python and the SimPy library to model complex network interactions and analyze performance metrics. The final strategy successfully achieved near-perfect fairness (Jain's Fairness Index > 0.97) and balanced channel occupancy (~45-50% per technology) in simulations, demonstrating a significant improvement over baseline models.

Data Analysis, System Simulation and +6 skills

Associated with University of Cape Town

Skills (48)

Python (Programming Language)

Machine Learning Internship at FlyRank AI

University of Cape Town

Final Year Thesis: Coexistence Strategy for 5G New Radio and Wi-Fi

Show all 6 details →

Wireless Communications

Engineering Intern at Mamoshe Investment Group (Pty) Ltd

Final Year Thesis: Coexistence Strategy for 5G New Radio and Wi-Fi

Show all →

Interests

Companies | Groups | Newsletters | Schools

Motorola Solutions
725,327 followers

Power Commerce
4,849 followers

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Causes

Education • Environment • Science and Technology • Poverty Alleviation • Social Services

Profile language

English

Public profile & URL

www.linkedin.com/in/brendon-mutema

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University of Cape Town

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Megchem

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Harare Institute of Technology

View



Government Policy Specialist in the Professional Organizations industry

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